

NAHWAN AL HIRZI

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Brebes, Central Java | Willing to relocate

Recent Data Science graduate skilled in Python, SQL, machine learning, and statistical modelling to solve open-ended business problems across customer acquisition, retention, and engagement. Experienced in building predictive models, customer segmentation, churn analysis, and interactive dashboards (Power BI, Google Data Studio, Google BigQuery) with strong data engineering foundations using Apache Airflow and Docker.

EDUCATION

Bachelor of Data Science, Telkom University Purwokerto

Graduated: January 2026

- GPA: 3.51/4.00
- Thesis: K-Means Optimization Using Whale Optimization Algorithm (WOA) to improve clustering quality and convergence performance compared to conventional K-Means.

WORK EXPERIENCE

Data Analyst Intern, Badan Pusat Statistik

July 2025 – August 2025

- Developed automated data collection and sentiment analysis workflows using Python, YouTube Data API, and Groq API to process 10,000+ YouTube comments and aggregate data from 23 news portals, enabling scalable text analytics and research reporting.

Laboratory Assistant, Telkom University Purwokerto

February 2025 – January 2026

- Delivered Business Intelligence, Database System, and Business Analytics practical sessions to 120+ students using Python, SQL, Power BI, and Google Data Studio to support end-to-end analytical workflows, dashboard development, and data-driven business reporting.

Machine Learning Cohort (MSIB) — Bangkit Academy

October 2024 – February 2025

- Developed machine learning models using Random Forest and TensorFlow/Keras on 120,000+ health records to support an ML-based stunting prevention application, achieving 79.97% test accuracy through predictive modelling, transfer learning, and model optimization techniques.

Census Data Processing Staff, Badan Pusat Statistik

September 2025 – November 2025

- Processed and validated census datasets for 200+ statistical units using QGIS and ArcGIS to improve spatial data accuracy, support database integration, and assist operational reporting analysis.

PROJECT

MLOps Bitcoin Pipeline

March 2026

- Built an automated ELT pipeline using Apache Airflow & Docker ingesting real-time Bitcoin data from CoinGecko API; implemented CI/CD with GitHub Actions, data versioning with DVC, and 7-day forecasting with Scikit-learn MLP Regressor.

Gen Z Finance Insight Dashboard

November 2025

- Analyzed financial behavior data from 1,652+ respondents across 20 provinces using Power BI to identify spending and literacy patterns through 10+ KPI dashboards and data-driven insight reporting.

Telco Customer Churn Prediction

July 2025

- Built predictive machine learning models using Python, SQL, and statistical analysis techniques on 7,043 customer records to identify churn behavior and support customer retention strategies, achieving ROC AUC 0.84 and Recall 78.8% through Extra Trees optimization and interactive Power BI reporting.

Customer Segmentation Hierarchical Clustering

April 2025

- Built multi-approach customer segmentation (Value-Based, Hybrid, Revenue-Based) using Agglomerative Hierarchical Clustering (Python, sklearn), optimized via manual grid search, achieving silhouette scores of 0.84–0.95 across models.

Time Series Analysis — AAPL Stock Price Forecasting

February 2025

- Forecasted Apple Inc. (AAPL) stock prices using ARIMA (R) on 713 trading-day records (Jan 2022–Nov 2024), conducting stationarity testing (ADF), ACF/PACF/EACF analysis across 33 candidate models, achieving MAPE of 1.28% with the best-fit ARIMA(0,1,2) model.

AWARDS

- 6th Place Winner, National Dashboard Analysis Tournament, UPN Veteran Jawa Timur November 2025
Competed against 49 teams nationwide in a data visualization competition, ranking 6th by delivering comprehensive business insights through an interactive Power BI dashboard.

CERTIFICATIONS

- Data Pipelines with TensorFlow Data Services (DeepLearning.AI) - Coursera
Credential ID: [Q7J2ZAAYL91H](#)
- Machine Learning Specialization (DeepLearning.AI) - Coursera
Credential ID: [0LI8AH1G9OTT](#)
- Crash Course on Python - Coursera
Credential ID: [6CSK2AV268EM](#)
- DeepLearning.AI TensorFlow Developer - Coursera
Credential ID: [1IRW2HVZ6RWN](#)
- Convolutional Neural Networks in TensorFlow - Coursera
Credential ID: [FH5DIF2QOAH1](#)
- Generative AI for Everyone - Coursera
Credential ID: [5YMCYMAAO316](#)
- Introduction SQL – MySkill

SKILLS

Language: Bahasa Indonesia (Native), English (Intermediate)

Languages & Query: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Matplotlib), SQL (CTE, Window Functions, Aggregation)

Machine Learning & Statistics: Predictive Modelling, Classification, Clustering, Time Series, Transfer Learning, Feature Engineering

Data Engineering: Apache Airflow, Docker, DVC, GitHub Actions, Git

Cloud & Database: Google BigQuery, Google Cloud Platform, PostgreSQL, MySQL

Visualization: Power BI, Google Data Studio

Geospatial: QGIS, ArcGIS